

Md Hafijur Rahman

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EDUCATION

The Pennsylvania State University

Ph.D. in Mechanical Engineering; CGPA: 4.00/4.00

Pennsylvania, USA

Jan 2023 – Present

University of Alberta

M.Sc. in Mechanical Engineering; CGPA: 4.00/4.00

Edmonton, Alberta, Canada

Jan 2021 – Dec 2022

Bangladesh University of Engineering and Technology

B.Sc. in Mechanical Engineering; CGPA: 3.81/4.00

Dhaka, Bangladesh

Jul 2014 – Oct 2018

GRADUATE COURSES

Physics of Radiation Damage, Manufacturing Methods in Microelectronics, Continuum Mechanics, Macro Fracture Mechanics, Fundamentals of Engineering Numerical Analysis, Applied Computational Intelligence for Engineers, Mechanics and Design of Composite Materials.

RESEARCH INTERESTS

My research focuses on **defect mitigation**, **radiation resilience**, and **microstructural modification** in materials and semiconductor devices. I specialize in low-temperature annealing using **Electron Wind Force (EWF)** to mitigate defects in **wide-bandgap (WBG) semiconductors** including **GaN** and **SiC**, as well as structural alloys including **FeCrAl** and **ZrTi**. My work integrates **EBSD**, **XRD**, **TEM**, **FIB**, **Raman spectroscopy**, electropulsing, in-situ characterization, and device-level electrical characterization. I also have a background in continuum mechanics, finite element analysis, molecular dynamics simulation, machine learning, and fracture mechanics.

RESEARCH EXPERIENCE

The Pennsylvania State University

Graduate Research Assistant – Haque Research Group

Pennsylvania, USA

Jan 2023 – Present

- **Low-Temperature Defect Engineering via Electron Wind Force:** Playing a leading role in NSF-funded research to develop EWF as an alternative to conventional high-temperature annealing. Designed electropulsing systems, implemented in-situ probing, and established protocols to mitigate defects in structural alloys and semiconductor devices.
- **Radiation Effects in Wide-Bandgap Semiconductors:** Investigated gamma and heavy-ion radiation effects in WBG semiconductor devices including GaN HEMTs, SiC MOSFETs, and Zener diodes, with emphasis on EWF-based defect mitigation as an alternative to conventional thermal annealing.
- **Restorative and Preemptive EWF Annealing of Semiconductor Devices:** Demonstrated room-temperature EWF rejuvenation of degraded GaN transistors and repeated recovery of radiation-damaged SiC MOSFETs. Developed preemptive EWF treatment strategies to reduce pre-existing defects and improve post-radiation resilience.
- **EWF-Driven Defect Mitigation in Structural Alloys:** Developed and applied low-temperature electropulsing protocols to FeCrAl and ZrTi alloys. Demonstrated elimination of low-angle grain boundary networks, grain boundary realignment, and microstructural modification confirmed by EBSD, XRD, Raman spectroscopy, and TEM.
- **In-situ TEM Evidence of Athermal Defect Recovery:** Performed in-situ TEM electropulsing experiments to capture real-time defect annihilation, supporting EWF as a room-temperature defect recovery mechanism.

University of Alberta

Graduate Research Assistant – Theoretical and Applied Mechanics Laboratory

Edmonton, Alberta, Canada

Jan 2021 – Dec 2022

- **Continuum Modeling of Fiber-Reinforced Composites:** Developed higher-gradient continuum models for elastic materials reinforced with extensible fibers and implemented the formulations using finite element analysis in FEniCS. This work resulted in two first-author peer-reviewed journal publications.

Bangladesh University of Engineering and Technology

Undergraduate Thesis with Dr. Mohammad Nasim Hasan

Dhaka, Bangladesh

Aug 2017 – Oct 2018

- **Molecular Dynamics Study of Condensation on Nanostructured Surfaces:** Conducted non-equilibrium molecular dynamics simulations to investigate argon vapor condensation on platinum nanostructured surfaces, including the effects of surface geometry, wettability, and wall temperature.

PUBLICATIONS (PEER-REVIEWED JOURNAL)

21. **Rahman, M.H.**, Wan, H.H., Ren, F., Wolfe, D.E., Haque, A. Room-Temperature Electrical Processing of Aluminum Nitride Heterojunction Diodes. *Journal of Physics: Materials*, **2026**. DOI: <https://doi.org/10.1088/2515-7639/ae6e74>
20. **Rahman, M.H.**, Liu, G., Long, J., Islam, M.Z., Mu, Y., Cao, D., Haque, A. Achieving Microstructural Enhancements at Sub-Zero Temperatures. *Materials Characterization*, **2026**, 232, 116016. DOI: <https://doi.org/10.1016/j.matchar.2026.116016>
19. **Rahman, M.H.**, Al-Mamun, N.S., Oh, H., Haque, A. Athermal Defect Recovery in High Entropy Alloys: Real-Time Evidence from In-situ TEM. *Scripta Materialia*, **2025**, 271, 117025. DOI: <https://doi.org/10.1016/j.scriptamat.2025.117025>
18. **Rahman, M.H.**, Al-Mamun, N.S., Stepanoff, S.P., Haque, A., Ren, F., Pearton, S.J., Wolfe, D.E. Room Temperature Rejuvenation Technology for Irradiated Gallium Nitride Transistors. *Advanced Materials Technologies*, **2025**, e00874. DOI: <https://doi.org/10.1002/admt.202500874>
17. **Rahman, M.H.**, Chavda, C., Al-Mamun, N.S., Stepanoff, S.P., Haque, A., Wolfe, D.E., Ren, F., Pearton, S.J. Repeated Rejuvenation of SiC MOSFETs for Unprecedented Ionizing Radiation Resilience. *APL Electronic Devices*, **2025**, 036103. DOI: <https://doi.org/10.1063/5.0266544>
16. **Rahman, M.H.**, Cooper, F., Crespillo, M.L., Hattar, K., Haque, A., Ren, F., Pearton, S., Wolfe, D. Improving Radiation Tolerance with Room Temperature Annealing of Pre-Existing Defects. *Applied Physics Express*, **2025**, 18, 17001. DOI: <https://doi.org/10.35848/1882-0786/adab50>
15. **Rahman, M.H.**, Rasel, M.A.J., Smyth, C.M., Waryoba, D., Haque, A. Radiation Damage Mitigation in FeCrAl Alloy at Sub-Recrystallization Temperatures. *Materials*, **2025**, 18, 124. DOI: <https://doi.org/10.3390/ma18010124>
14. **Rahman, M.H.**, Chavda, C., Warner, L., Stafford, S., Carvajal, J., Haque, A., Ren, F., Pearton, S., Wolfe, D.E. Room Temperature Annealing of Gamma Radiation Damage in Zener Diodes. *ECS Journal of Solid State Science and Technology*, **2025**, 14, 025003. DOI: <https://doi.org/10.1149/2162-8777/adb685>
13. **Rahman, M.H.**, Oh, H., Waryoba, D., Haque, A. Microstructural Modification and Enhanced Mechanical Properties in Zr50–Ti50 Alloy via Low Temperature Electron Wind Force Annealing. *Materials Characterization*, **2024**, 215, 114188. DOI: <https://doi.org/10.1016/j.matchar.2024.114188>
12. **Rahman, M.H.**, Todaro, S., Waryoba, D., Haque, A. Synergistic Thermal and Electron Wind Force-Assisted Annealing for Extremely High-Density Defect Mitigation. *Materials*, **2024**, 17, 3188. DOI: <https://doi.org/10.3390/ma17133188>
11. **Rahman, M.H.**, Todaro, S., Warner, L., Waryoba, D., Haque, A. Elimination of Low-Angle Grain Boundary Networks in FeCrAl Alloys with the Electron Wind Force at a Low Temperature. *Metals*, **2024**, 14, 331. DOI: <https://doi.org/10.3390/met14030331>
10. **Rahman, M.H.**, Glavin, N., Haque, A., Ren, F., Pearton, S.J. Effect of High Current Density Pulses on Performance Enhancement of Optoelectronic Devices. *ECS Journal of Solid State Science and Technology*, **2024**, 13, 25003. DOI: <https://doi.org/10.1149/2162-8777/ad28c8>

9. **Rahman, M.H.**, Warner, L., Bae, J., Kim, J., Haque, A., Ren, F., Pearton, S.J., Wolfe, D.E. Improving Radiation Resilience of Zener Diodes Through Preemptive and Restorative Electron Wind Force Annealing. *Physica Scripta*, **2024**, 100, 015904. DOI: <https://doi.org/10.1088/1402-4896/ad943b>
8. **Rahman, M.H.**, Al-Mamun, N.S., Glavin, N., Haque, A., Ren, F., Pearton, S., Wolfe, D.E. Rejuvenation of Degraded Zener Diodes with the Electron Wind Force. *Applied Physics Express*, **2024**, 17, 47001. DOI: <https://doi.org/10.35848/1882-0786/ad379c>
7. **Rahman, M.H.**, Oh, H., Waryoba, D., Haque, A. Room Temperature Control of Grain Orientation via Directionally Modulated Current Pulses. *Materials Research Express*, **2023**, 10, 116521. DOI: <https://doi.org/10.1088/2053-1591/ad0b56>
6. **Rahman, M.H.**, Yang, S., Kim, C. II. A Third Gradient-Based Continuum Model for the Mechanics of Continua Reinforced with Extensible Bidirectional Fibers Resistant to Flexure. *Continuum Mechanics and Thermodynamics*, **2023**, 35, 563–593. DOI: <https://doi.org/10.1007/s00161-023-01198-9>
5. **Rahman, M.H.**, Islam, S., Yang, S., Kim, C. II. A Shear Lag Theory Integrated with Second Strain Gradient Continuum Model for the Composite Reinforced with Extensible Nano-Fibers. *Acta Mechanica*, **2023**, 234, 4269–4296. DOI: <https://doi.org/10.1007/s00707-023-03612-4>
4. Chavda, C, **Rahman, M.H.**, Warner, L., Yeager, J.L., Haque, A., Wolfe, D.E., Ren, F., Pearton, S.J. Cold annealing of gamma irradiated high-voltage SiC MOSFETs. *APL Electronic Devices*, **2026**, 2, 016116. DOI: <https://doi.org/10.1063/5.0304589>
3. Thomas, M.P., Schoell, R., Rasel, M.A.J, **Rahman, M.H.**, Kuo, W., Watt, J., House, S., Hattar, K., Windes, W., Haque, A. An In Situ Transmission Electron Microscopy Study on the Synergistic Effects of Au-Ion Irradiation and High Temperature on Nuclear Graphite Microstructure. *Materials Research Express*, **2024**, 11, 45601. DOI: <https://doi.org/10.1088/2053-1591/ad37a6>
2. Thomas, M.P., Schoell, R., Rasel, M.A.J, **Rahman, M.H.**, Kuo, W., Watt, J., House, S., Hattar, K., Windes, W., Haque, A. Lateral NiO/AlN Heterojunction Rectifiers with Breakdown Voltage >11 kV. *ECS Advances*, **2024**, 3, 33502. DOI: <https://doi.org/10.1149/2754-2734/ad7867>
1. Liu, G., Oh, H, **Rahman, M.H.**, Du, J., Windes, W., Haque, A. Low-Temperature Annealing of Nanoscale Defects in Polycrystalline Graphite. *Carbon*, **2024**, 10(3), 76. DOI: <https://doi.org/10.3390/c10030076>

MANUSCRIPTS IN PREPARATION

- **Rahman, M.H.**, et al. (with the group of Prof. Houqiang Fu, Arizona State University). EWF Processing of AlN Heterojunction Diodes: Reduction of Sheet and Contact Resistance. *Manuscript in preparation*.

CONFERENCE PRESENTATIONS AND INVITED TALKS

6. **Rahman, M.H. INVITED POSTER:** Reliability of Ga₂O₃ Schottky Barrier Diodes. *International Workshop on Gallium Oxide and Related Materials (IWGO 2026)*, University of Maryland, College Park, MD, USA, **August 2–7, 2026**. Invited Poster Session, Program No. IWGO-MoPI-1 (scheduled August 3, 2026).
5. **Rahman, M.H.** Rejuvenating Materials and Electronics: A Non-Thermal Approach to Defect Mitigation. *Materials Science & Technology (MS&T) Technical Meeting and Exhibition, Processing and Performance of Materials Using Microwaves, Electric and Magnetic Fields, Ultrasound, Lasers, and Mechanical Work – Rustum Roy Symposium*, **October 4–7, 2026**, Pittsburgh, PA, USA. Oral presentation (abstract accepted).
4. **Rahman, M.H.** Enhancing Radiation Resilience of Wide-Bandgap Semiconductors and Alloys via Electron Wind Force Annealing. *Materials Science & Technology (MS&T) Technical Meeting and Exhibition, Processing and Performance of Materials Using Microwaves, Electric and Magnetic Fields, Ultrasound, Lasers, and Mechanical Work – Rustum Roy Symposium*, **2025**, Columbus, OH, USA. Oral presentation.
3. **Rahman, M.H.** Defect and Microstructure Control of Materials Using the Electron Wind Force. *Materials Science & Technology (MS&T) Technical Meeting and Exhibition, Processing and Performance of Materials*

Using Microwaves, Electric and Magnetic Fields, Ultrasound, Lasers, and Mechanical Work – Rustum Roy Symposium, 2024, Pittsburgh, PA, USA. Oral presentation.

2. **Rahman, M.H.**, Chavda, C., Al-Mamun, N., Stepanoff, S., Haque, A., Wolfe, D., Ren, F., Pearton, S. (Invited) Electron Wind Force Effects on Defects in Wide Bandgap Electronics. *248th ECS Meeting*, Chicago, IL, October 12-16, 2025. Volume MA2025-02, H03: *Gallium Nitride and Silicon Carbide Power Technologies* 15. Invited oral presentation (delivered by A. Haque). DOI: 10.1149/MA2025-02351707mtgabs
1. **Rahman, M.H.** Defect and Microstructure Control of Materials Using the Electron Wind Force. *Low Carbon Energy Systems (LCES) Talk Series, College of Engineering, The Pennsylvania State University, 2024*. Invited talk.

MENTORSHIP AND STUDENT SUPERVISION

Over the course of my Ph.D., I have supervised six undergraduate researchers and trained four junior graduate students. Several mentees have co-authored multiple papers and gone on to graduate programs (e.g., Purdue University) or received research awards.

Undergraduate Researchers

- **Hajin Oh** *Spring 2023 – Fall 2023*
 - **Project:** Low-temperature microstructural control in alloys via Electron Wind Force (EWF).
 - **Publications:** 3 published journal papers, 1 under review.
 - **Achievement:** Admitted to Ph.D. program, **Purdue University**.
- **Sarah Todaro** *Spring 2024 – Fall 2024*
 - **Project:** Rapid defect mitigation in FeCrAl alloy using EWF.
 - **Publications:** 2 journal papers.
- **Felix Cooper** *Fall 2024 – Spring 2025*
 - **Project:** Preemptive EWF treatment of Zener diodes for radiation resilience.
 - **Publications:** 1 journal paper.
- **Luke Warner** *Spring 2024 – Summer 2025*
 - **Project:** Radiation effects in WBG semiconductors and EWF recovery.
 - **Publications:** 3 journal papers.
 - **Recognition:** 2025 Dr. John P. Karidis Department Head's Award (ME).
- **Patel Chaitanya** *Spring 2025 – Present*
 - **Project:** Sub-zero direct-current processing of metals.
- **Lorenzo Adrian Zamel** *Spring 2025 – Present*
 - **Project:** Enhancing harsh-environment resilience of GaN transistors via EWF rejuvenation.

Junior Graduate Students (Training & Onboarding)

- Mentored **Adnan Mahathir**, **Fahim Mahtab Abir**, **Xinwei Wang**, and **Felix Nikhil Kumar (M.S.)** in early-stage research, covering safety/SOPs, electropulsing hardware, EBSD/XRD/TEM workflows, data analysis, and manuscript preparation.

HONORS, AWARDS AND LEADERSHIP

- **Harry G. Miller Fellowship in Engineering**, Penn State College of Engineering, Fall 2025. Awarded for interdisciplinary research and travel support to present at MS&T25.
- **Invited Poster Presenter**, International Workshop on Gallium Oxide and Related Materials (IWGO 2026), University of Maryland, College Park, MD, August 2026.
- **Vice President**, Bangladesh Student Association, The Pennsylvania State University, 2025.

- **Dean's List Award**, Department of Mechanical Engineering, Bangladesh University of Engineering and Technology, 2016–2018.
- **University Merit Scholarship**, Bangladesh University of Engineering and Technology, 2015–2018.

TEACHING EXPERIENCE

The Pennsylvania State University	Pennsylvania, USA
<i>Graduate Teaching Assistant – ME 460: Mechanical Design II</i>	<i>Jan 2023 – May 2023</i>
University of Alberta	Edmonton, Alberta, Canada
<i>Graduate Teaching Assistant – MEC E 301 and EN PH 131</i>	<i>Aug 2021 – Dec 2022</i>
Bangladesh Army University of Science and Technology	Saidpur, Bangladesh
<i>Lecturer/Instructor – Department of Mechanical Engineering</i>	<i>Nov 2019 – Dec 2020</i>

KEY SKILLS

Materials Characterization:

SEM (Verios G4 UC; Apreo S); EBSD (Aztec); FIB (Scios 2; Helios 660); TEM/STEM (Talos F200X); EDS (Aztec for SEM; Velox for TEM); XRD (Empyrean); Raman spectroscopy (Horiba Soleil); Optical Profilometer.

Mechanical & Thermal Testing:

Nanoindentation (Bruker Hysitron TI-980); Tensile testing.

Data, Modeling & Programming:

Python; MATLAB; Finite Element Analysis (FEniCS); SRIM; LAMMPS.

Electrical Characterization:

IV/CV measurement: SMUs (Keithley 2636B, 4200, 2400); Probe station; Pulse meter (Keithley 2601B); DC power supply.

Cleanroom Process Exposure:

Photolithography; Deep Reactive-Ion Etching (DRIE); Wet etching.

Tools & Software:

OriginPro; ImageJ/Fiji; AZtecCrystal; LabSpec 6; Jade; LaTeX.